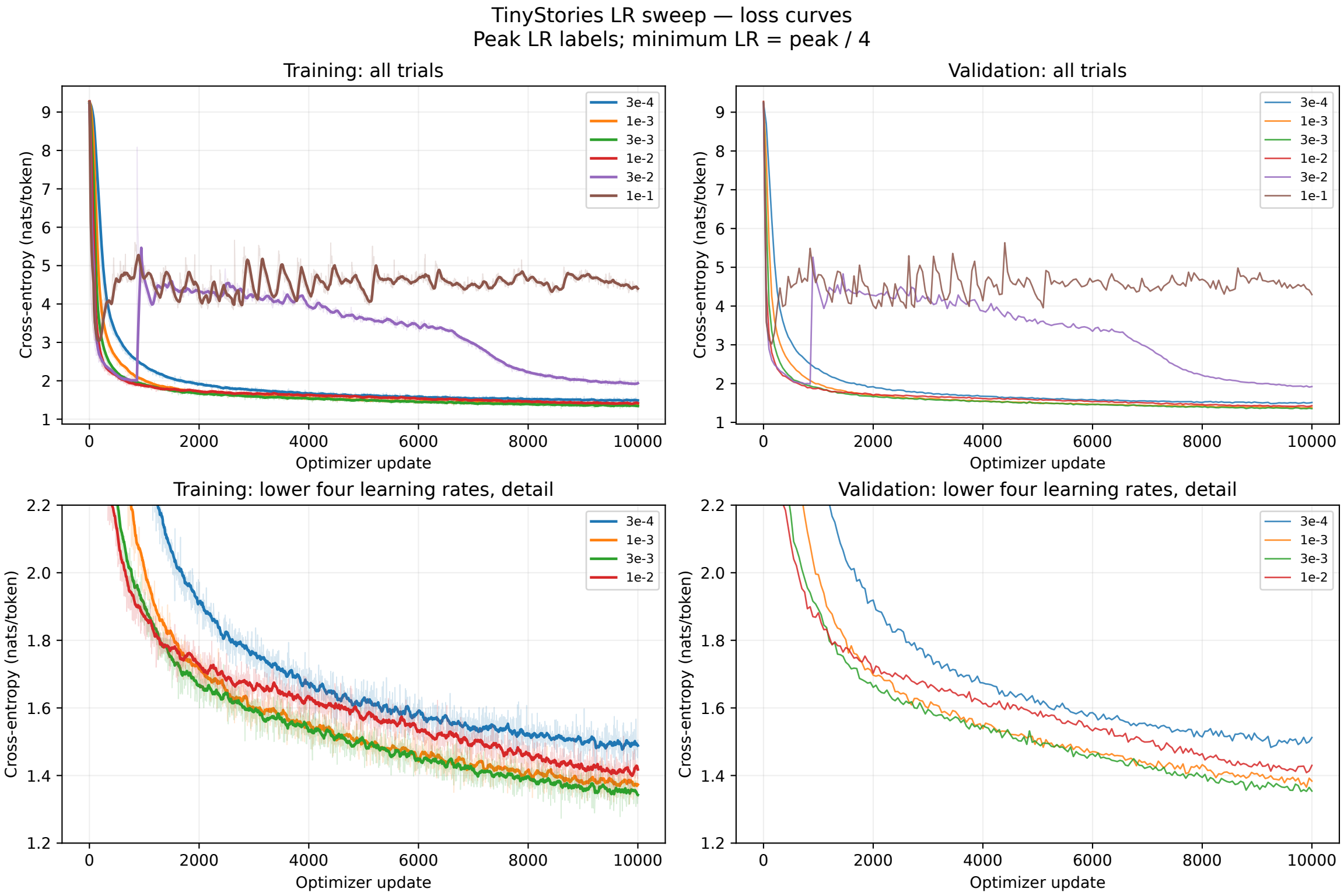
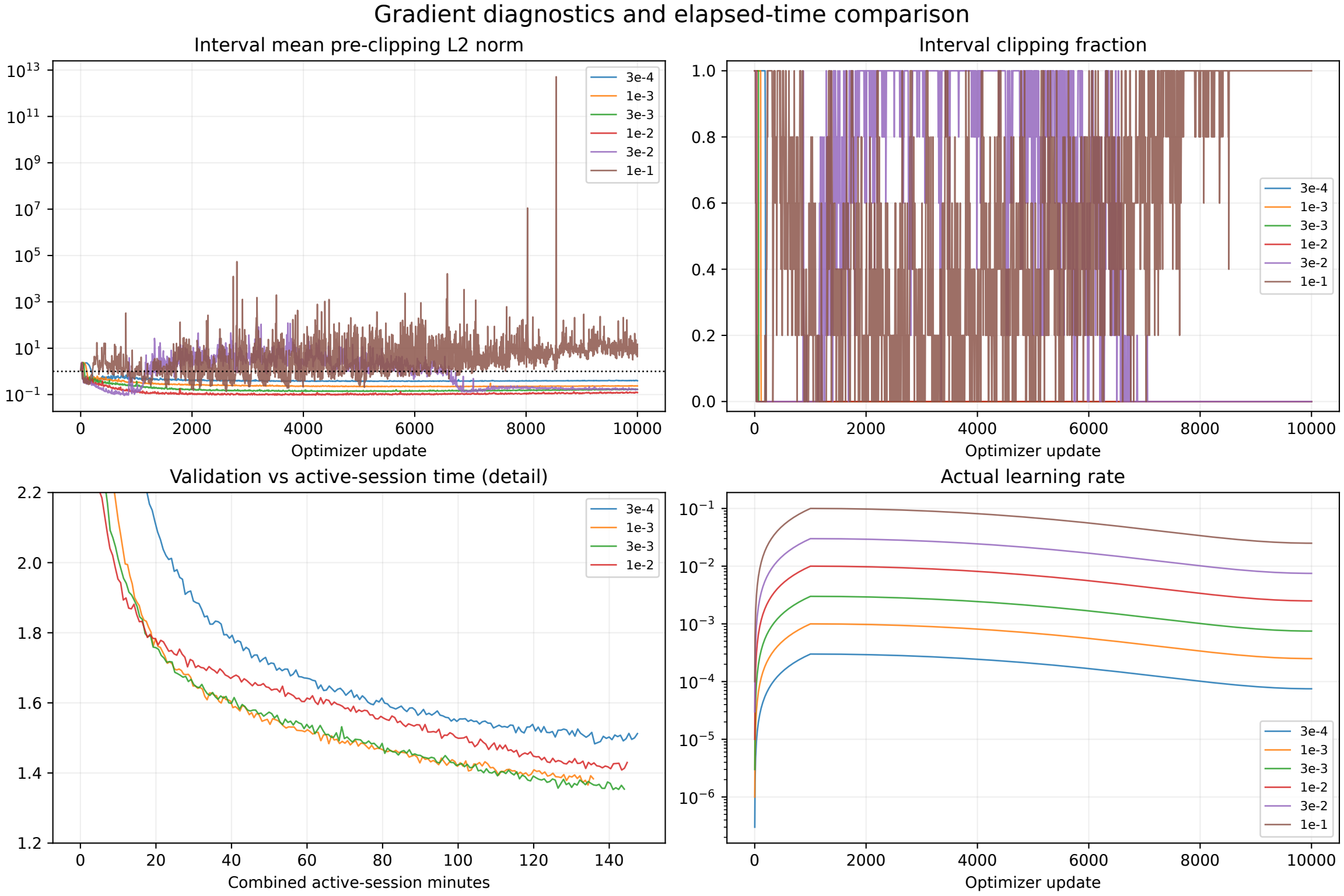

TinyStories LR sweep — loss curves
Peak LR labels; minimum LR = peak / 4
Training: all trials
Validation: all trials
Training: lower four learning rates, detail
Validation: lower four learning rates, detail
Cross-entropy (nats/token)
Optimizer update
3e-4
1e-3
3e-3
1e-2
3e-2
1e-1


Gradient diagnostics and elapsed-time comparison
Interval mean pre-clipping L2 norm
Optimizer update
Interval clipping fraction
Optimizer update
Validation vs active-session time (detail)
Combined active-session minutes
Actual learning rate
Optimizer update
3e-4
1e-3
3e-3
1e-2
3e-2
1e-1

Export notes and source runs

Exported UTC: 2026-11-06T16:18:01.100495+00:00

Training: raw logged batches (alpha 0.18) plus trailing mean of 16 samples (~80 updates).
Early means use available samples. Validation averages 16 batches and is not smoothed.
Training samples span changing model weights; matching batch counts does not ensure equal noise.
Clipped detail panels are explicitly zoomed; full loss curves appear on page 1.
1e-3 joins two sessions by optimizer step. Replayed updates use the newer session.
Time sums session timers, excludes downtime, and retains elapsed work discarded on restart.
Validation time is estimated from W&B timestamps and neighboring training timer samples.
Session timers exclude process startup; final validation time is estimated, not precisely instrumented.
All six trials reached 10,000 updates. No multi-seed replication.
Architecture: 4 layers, width 512, FFN 1344, 16 heads, context 256, vocabulary 10,000.
Batch 128; 10,000 updates; 1,000 warmup updates; cosine decay to step 10,000.
AdamW: beta=(0.9, 0.999), weight decay=0.1; clipping threshold=1; float32 on A10.

3e-4 | 2iso2lvi | finished | last update 10000
1e-3 | 2a9winki | failed | last update 3816
1e-3 | y8519qgq | finished | last update 10000
3e-3 | 7arns66f | finished | last update 10000
1e-2 | ihk28xje | finished | last update 10000
3e-2 | bxd9y7lz | finished | last update 10000
1e-1 | bx0y6qc7 | finished | last update 10000